School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, Ettimadai, Coimbatore

22AIE305 — Introduction to Cloud Computing

Course Project Report

DiagramForge

A Cloud-Native DevSecOps Platform for
AI-Assisted Diagram Generation

Architecture, Implementation, and CI/CD Pipeline Report

Team D11

Name	Roll Number
Mahakisore M	CB.SC.U4AIE24333
Yashwanth B	CB.SC.U4AIE24360

Under the guidance of
Dr. Keerthika

October 14, 2026

Abstract

DiagramForge (repository name `Diagram-Ops`) is a cloud-native learning platform built around a deliberately small application: a user describes a system in natural language, a large language model (LLM) converts that description into Mermaid diagram syntax, and the browser renders it. The application is intentionally minimal so that the engineering effort can concentrate on the platform that surrounds it — containerisation, infrastructure as code, Kubernetes orchestration, a security-gated CI/CD pipeline, and (in later phases) observability, progressive delivery, and disaster recovery. This report documents the system as an industry-style architecture and implementation record: functional and non-functional requirements; the application's internal design (context, use case, component, data, control-flow, and state views); the infrastructure topology on AWS; and, in particular detail, the Jenkins-based DevSecOps pipeline that gates every change behind static analysis (SonarQube), software-composition analysis (OWASP Dependency-Check), and container image scanning (Trivy) before an image may reach the container registry. Real defects found while hardening that pipeline are documented as case studies, including a false-positive CVE caused by an NVD CPE vendor/product collision between the MongoDB server and its Node.js driver, and a class of image vulnerabilities caused by an unused `npm` CLI left in the runtime container. The report also records the first live deployment of the application to Amazon EKS through a Helm chart, including the cluster-level gaps it exposed (a missing default storage class, missing load-balancer IAM permissions, unlabelled subnets and an absent metrics server), the proof that database content survives deletion of the database pod, and the move to GitOps: ArgoCD reconciles the cluster from a Git branch that the pipeline updates, and Prometheus and Grafana report on the running service.

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1 Introduction

1.1 Motivation

Most student DevOps projects struggle for one of two opposite reasons: the application is either too trivial to justify a real platform, or too complex to finish. DiagramForge resolves this tension deliberately by keeping the *application* small — three persisted entities (users, diagrams, provider usage) and a single LLM-backed generation endpoint — while growing the *platform* around it, phase by phase, until its operational practices resemble what a small production team would actually run: infrastructure as code, a security-gated pipeline, GitOps delivery, observability, and explicit cost discipline.

1.2 Objectives

- Build a working three-tier application (React, Express, MongoDB) with real authentication, real LLM integration, and automated tests.
- Provision every piece of cloud infrastructure through Terraform, never through manual console operations.
- Enforce a non-negotiable quality bar before any container image reaches the registry: static analysis, dependency vulnerability scanning, and image vulnerability scanning, each able to fail the build independently.
- Run the application on Kubernetes (Amazon EKS), delivered through GitOps rather than direct `kubectl` access by a human.
- Treat cost as a first-class constraint — every phase's monthly cost is known and budgeted in advance.

1.3 Scope and Guiding Principle

The project's own guiding principle is deliberately blunt: *"Keep the app simple. Make the platform serious."* This report follows the same split. Section 4 covers the application's internal design; Section 5 covers the platform, which is the actual subject of the project.

2 Requirements Engineering

2.1 Actors

Actor	Description
Guest	An unauthenticated visitor who may register or log in.
Registered User	An authenticated user who generates, saves, edits, and exports diagrams.
LLM Provider (system)	Groq (primary) or Anthropic Claude (fallback); external systems the backend calls to convert text to Mermaid syntax.
Scheduler (system)	Triggers scheduled operational jobs such as nightly database backups.
Operator / SRE	Interacts with the platform only through Git commits and monitoring dashboards — never directly with the running application.

Table 1: System actors

2.2 Functional Requirements

The full specification enumerates requirements as `FR-<area><n>`, prioritised with RFC 2119 keywords (**MUST**, **SHOULD**, **MAY**). Table 2 summarises each area and calls out the individual requirements that most directly shaped the design in Section 4.

Area	Count	Architecturally significant requirements
Authentication (`FR-A*`)	9	Registration and login with bcrypt password hashing (cost $\geq$ 12); JWT-protected routes return `401` when the token is missing, invalid, or expired.
Generation (`FR-G*`)	9	`FR-G4`: on a syntax/type mismatch, retry *once*, using the *same* provider that responded (a "sticky retry") — a formatting glitch never escalates to the paid provider. `FR-G5`: automatic failover to the secondary provider. `FR-G6`: a hard daily call cap on the paid provider, enforced *before* the call. `FR-G7`: generation never persists anything; the user must explicitly save.
Diagram management (`FR-D*`)	8	`FR-D7`: a user may only read, update, or delete their own diagrams; any attempt on another user's diagram returns `404` rather than `403`, so the response never confirms that the resource exists.
Export (`FR-E*`)	3	SVG download, PNG download, copy raw Mermaid source.
Operational (`FR-O*`)	6	`FR-O1`/`FR-O2`: separate liveness (`/healthz`) and readiness (`/readyz`) probes, so a transient database blip cannot make Kubernetes restart every pod at once. `FR-O6`: graceful shutdown on `SIGTERM`.

Table 2: Functional requirement areas

2.3 Non-Functional Requirements

Category	Key requirement	Verification
Performance (`NFR-P*`)	95% of generations complete within 10 s; 95% of non-LLM requests within 300 ms.	Prometheus latency histograms (Phase 9).
Availability (`NFR-A*`)	99% non-5xx SLO over a rolling 7 days; RTO $\leq$ 45 min, RPO $\leq$ 24 h.	Recording rules, burn-rate alerts, timed restore drills.
Security (`NFR-S*`)	No secret in Git, image layers, or logs; non-root containers. **A build fails on any dependency with CVSS $\geq$ 8, or any fixable HIGH/CRITICAL image vulnerability** (`NFR-S5`).	OWASP Dependency-Check and Trivy gates (Phase 6, complete).
Scalability (`NFR-SC*`)	Backend scales from 2 to 6 pods above 70% average CPU in the production values; stateless backend.	HPA (rendered and verified in the chart; load test planned).
Observability (`NFR-O*`)	Per-route rate, errors, and latency visible; logs survive restarts and are queryable by request ID.	Grafana, Loki (Phases 9, 15).
Cost (`NFR-C*`)	Total AWS spend within the account's credit allowance, with disciplined teardown.	AWS Budgets alerts at \$25, \$50 and \$75; spend visible per session.

Table 3: Non-functional requirement categories

3 System Architecture

3.1 System Context

Figure 1 places the system in its operating environment. The Operator never touches the running application directly: they interact only through Git, which drives the pipeline, and through dashboards, which are read-only. This separation is the architectural point of GitOps — the cluster's desired state lives in a repository, not in anyone's memory of commands they once ran against it. The dashed edge to the Anthropic API is the failover path; it is exercised rarely by design, which is exactly why it has a dedicated automated test.

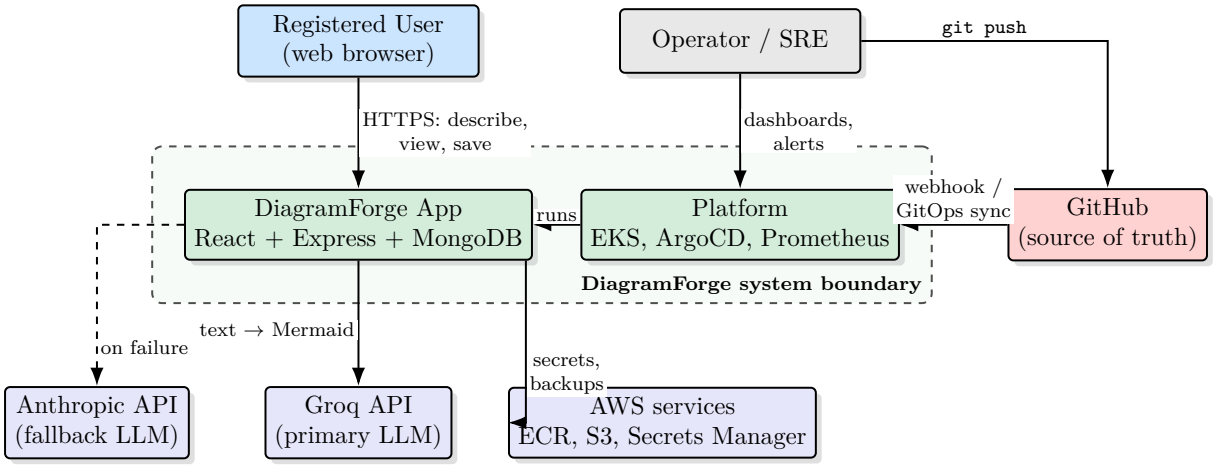


Figure 1: System context diagram

3.2 High-Level Runtime Architecture

Figure 2 shows the target runtime topology on Amazon EKS. The cluster itself is provisioned and verified (Phase 4); the application workloads are deployed onto it through Helm and ArgoCD in Phases 7–8, and today the same three-tier stack runs locally under Docker Compose. The frontend pod runs nginx, which serves the built React bundle and reverse-proxies `/api/*` to the backend service. The frontend calls `/api/...` as a *relative* path, so the browser only ever talks to a single origin: no CORS configuration is needed, and no API URL is baked into the bundle. This matters because Vite resolves environment variables at *build* time, not run time — a baked-in absolute URL would tie one image to one environment, which is the most common way this class of architecture breaks.

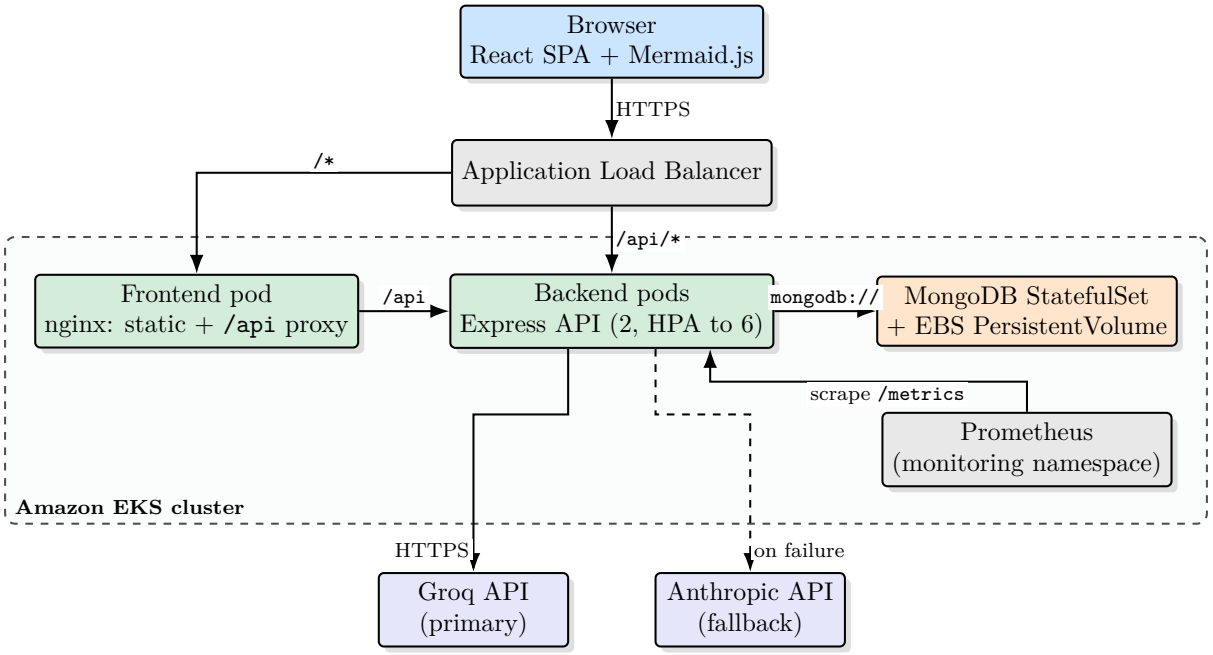


Figure 2: High-level runtime architecture on Amazon EKS

3.3 Layered Backend Component Architecture

The backend follows a strict layering discipline (Figure 3): controllers are thin and never touch models directly, all business logic lives in services, and the choice of LLM provider is hidden behind an interface. `ILLMProvider` is the architecturally important abstraction — `ProviderRegistry` depends on the *interface*, never on `GroqProvider` or `ClaudeProvider` directly (Dependency Inversion). That buys three concrete properties: adding a third provider is one new class and no other change; the whole test suite runs against a mocked interface with zero API calls and zero cost; and swapping which provider is primary is a single environment variable.

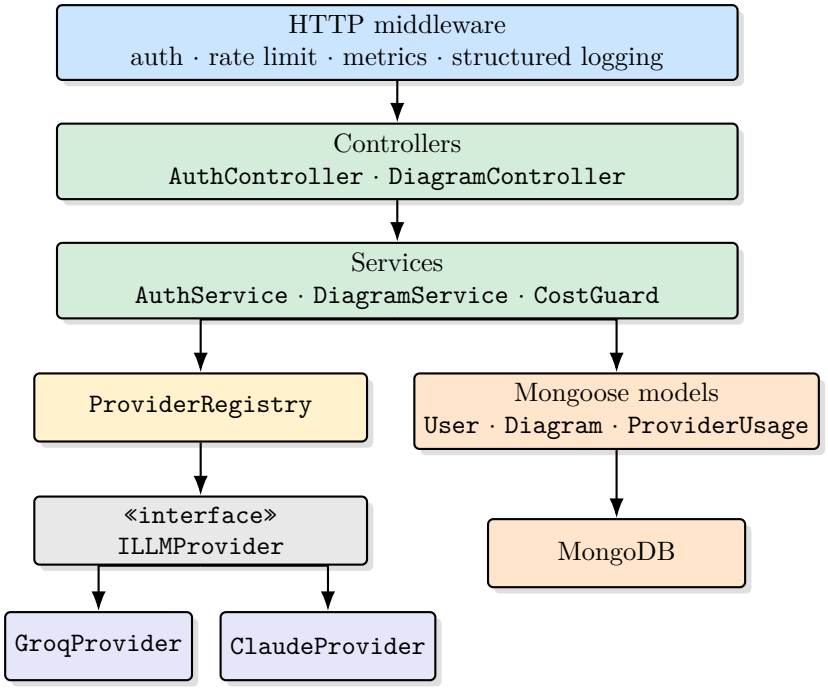


Figure 3: Layered backend component architecture

3.4 Data Model

MongoDB is a document database with no engine-enforced foreign keys, so the entity–relationship view in Figure 4 describes the *logical* model enforced at the Mongoose layer. Referential integrity — for example, removing a user's diagrams when the user is deleted — is therefore the application's responsibility, not the database's. `PROVIDER_USAGE` has no relationship line on purpose: it is a global daily aggregate, not a domain entity, and its unique compound index on `(date, provider)` is what makes the daily cap safe under concurrent requests.

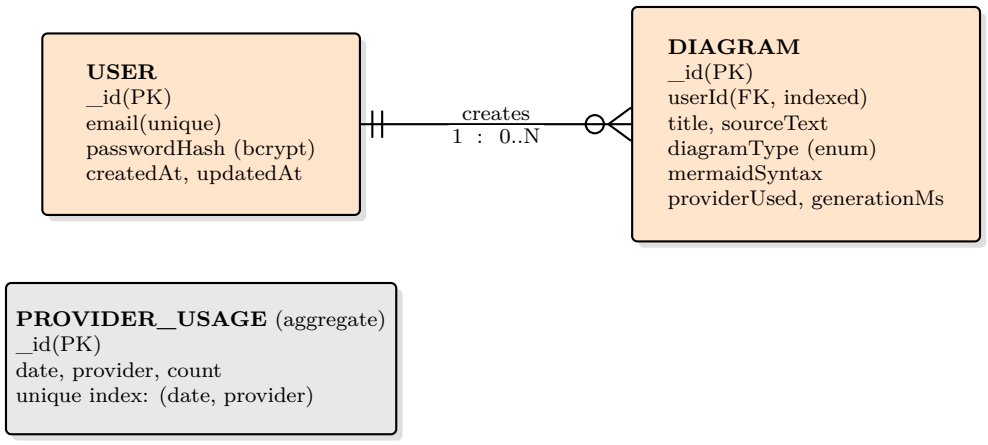


Figure 4: Entity–relationship model (crow's-foot notation, logical model enforced by Mongoose)

The compound index `{userId: 1, createdAt: -1}` on `DIAGRAM` exists because MongoDB can use a compound index only on a *prefix* of its fields. Every list query filters by `userId` and then sorts by `createdAt`, so `userId` must come first; reversed, the index would be useless for the most common query in the system.

4 Application Behaviour

4.1 Use Case Diagram

In Figure 5, `«include»` means the base use case *always* performs the included one — generating, viewing, and manually editing all necessarily render. `«extend»` means the extension happens *conditionally*: failover extends generation only when the primary provider actually fails.

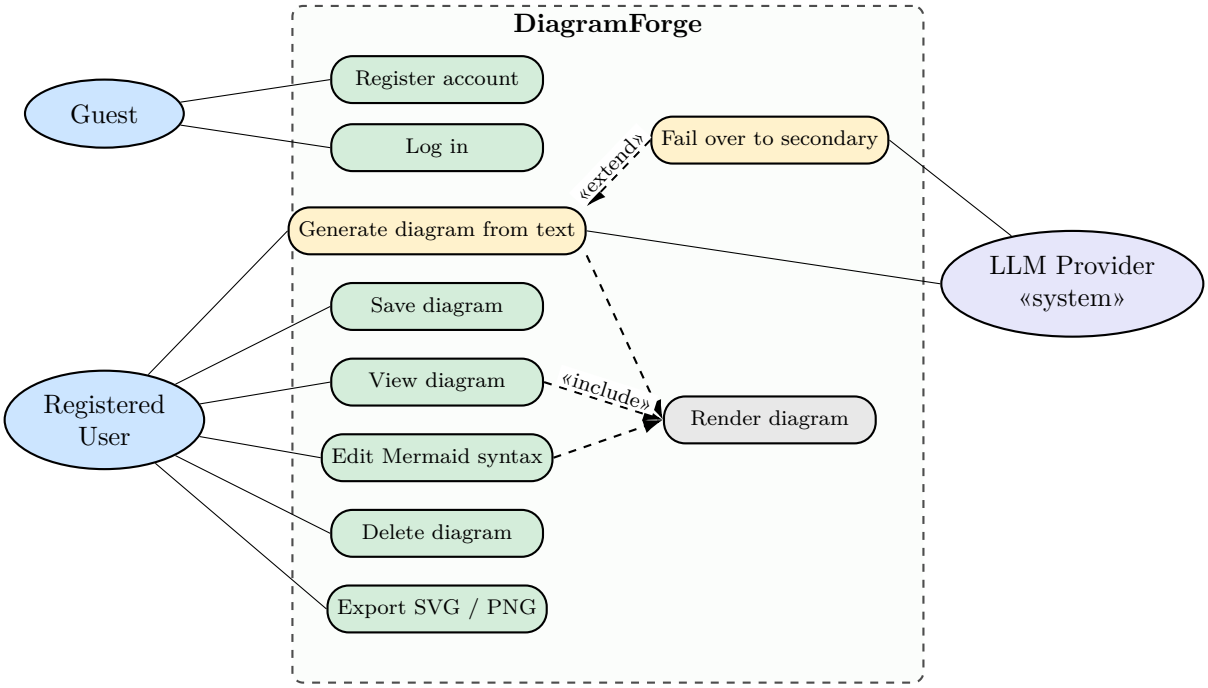


Figure 5: Use case diagram

4.2 Control Flow: Generation, Failover, and Cost Guard

Figure 6 is the most operationally important flow in the system, because it encodes the cost-control logic behind `NFR-C2`. Three invariants hold by construction, and each was checked against the implementation:

1. **Validation happens before any paid call**, so an oversized input costs nothing.
2. **The retry can happen at most once.** The "already retried?" gate routes to `502` on the second pass, so there is no loop and no unbounded spend.
3. **The daily cap is checked *before* the fallback call, never after.** A cap enforced after the money is spent is not a cap — it is a report.

4.3 Diagram Lifecycle

Figure 7 shows the lifecycle of a diagram from the user's point of view. The transition worth highlighting is `Rendered` → `Composing`: discarding a result is a first-class, free action, because generation never persists (`FR-G7`) — a bad draft leaves no orphaned row to clean up. `SyntaxError` is likewise contained: a Mermaid parse failure renders an inline error, and the page remains usable (`FR-D8`).

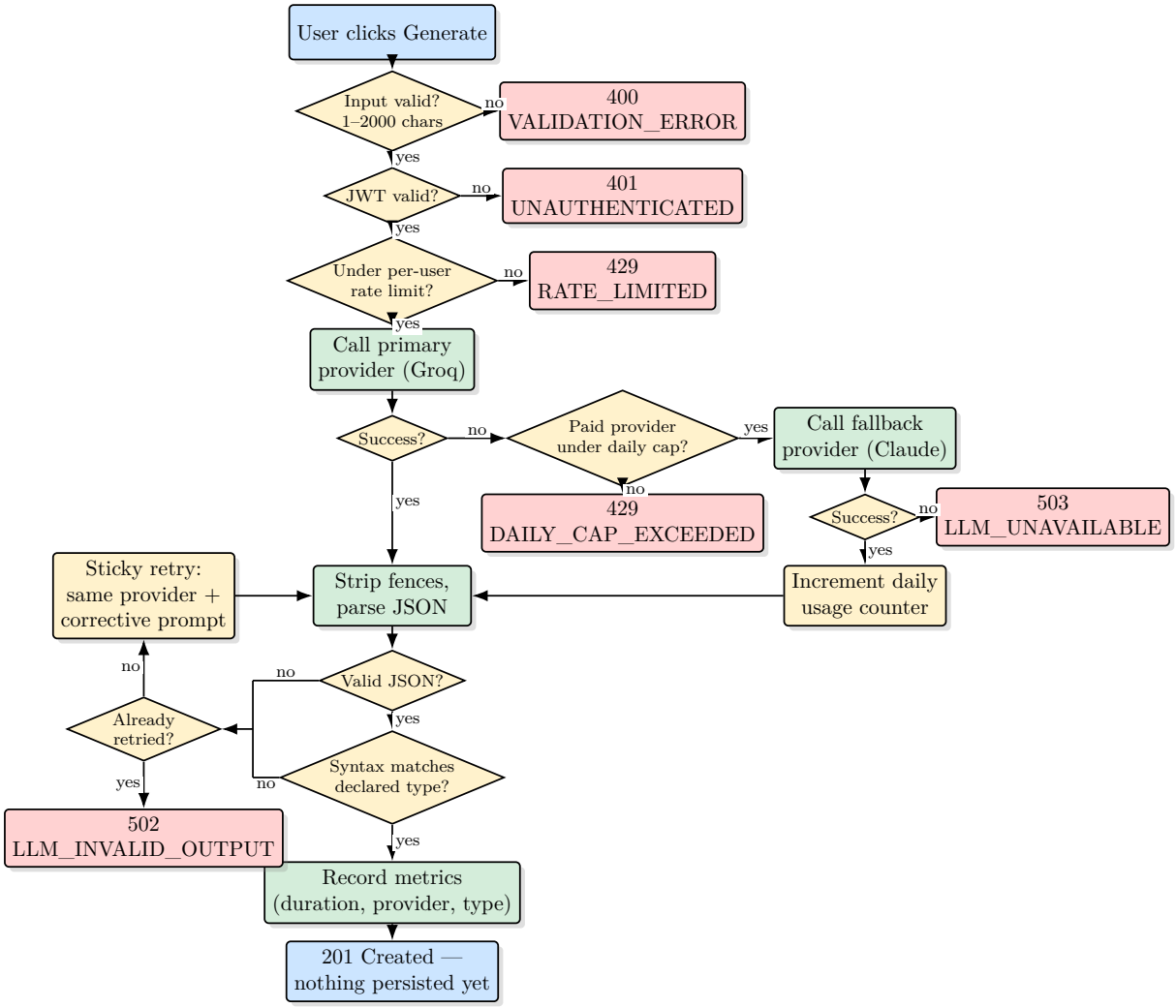


Figure 6: Generation, failover, and cost-guard control flow

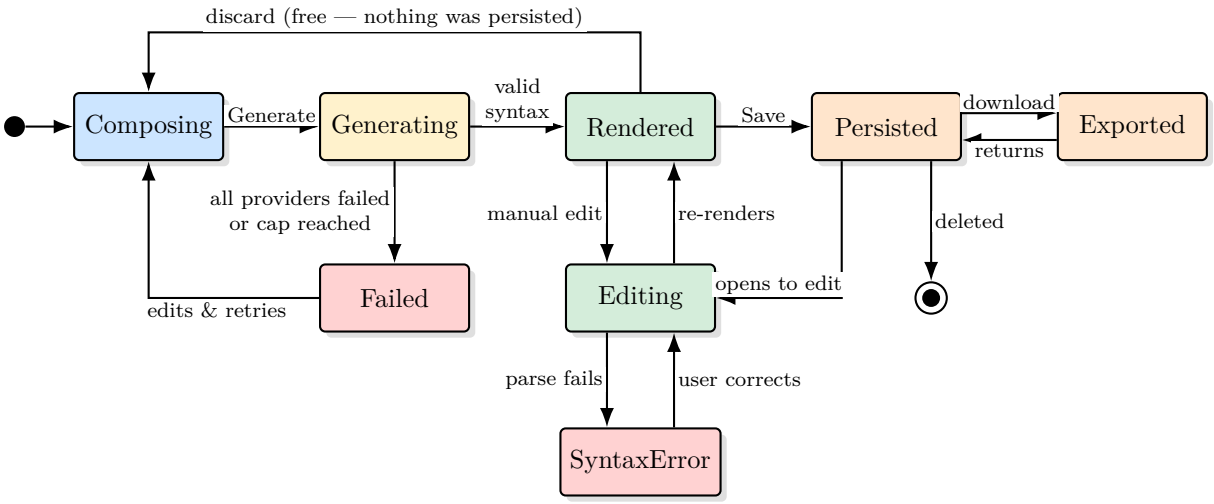


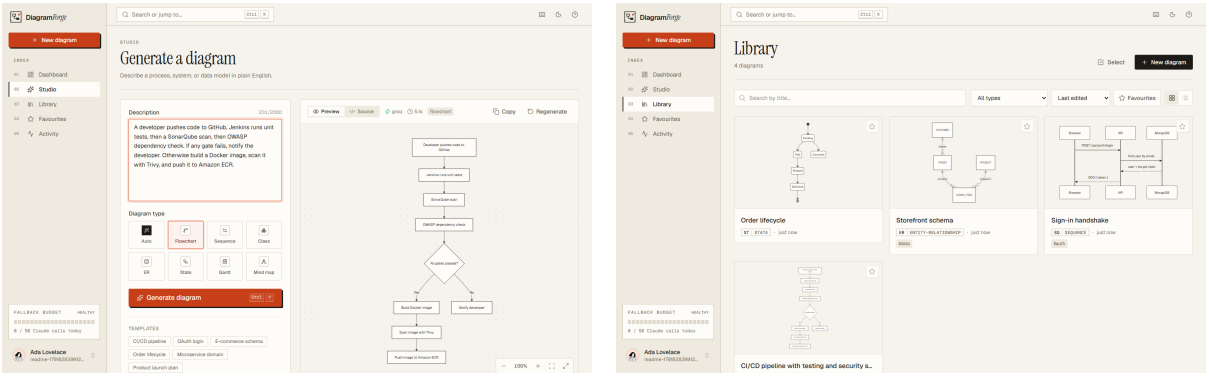
Figure 7: Diagram lifecycle state diagram

4.4 Product Enhancements and User Interface

Beyond the minimal generate–render–save loop of the original scope, the application was extended to the feature set expected of a multi-user product:

- **Library management:** search, filtering by diagram type and tag, favourites, duplication, sorting and pagination, with the last twenty versions of every diagram retained on edit.
- **Sharing:** revocable public links using unguessable 24-character tokens; shared pages are read-only, expose only render fields, and are rate-limited separately.
- **Account lifecycle:** profile and illustrated avatars rendered client-side, password change, a full machine-readable *data export*, and account deletion that cascades to the user's diagrams and audit events.
- **Audit log:** sign-ins, saves, shares and deletions recorded best-effort and expired automatically after ninety days by a MongoDB TTL index.
- **Interface:** a drafting-table design language with a light "paper" and a dark "cyanotype" theme, a command palette, keyboard shortcuts and a responsive layout down to phone width.

Figure 8 shows the studio, where a plain-English description produces a rendered flowchart in a few seconds, and the library view.



(a) Studio: generated CI/CD flowchart (b) Library: search, tags and favourites

Figure 8: The DiagramForge interface (light theme)

Avatars deserve a note because they show a privacy-minded design choice. The profile page (Figure 9) offers four illustration styles, six candidate faces per page and a shuffle control. Only

the recipe `{style, seed}` is stored on the user and validated against a fixed list on the server (`PATCH /auth/me`); the image itself is drawn in the browser by the DiceBear library. Nothing is uploaded, no image hosting is needed and no third-party avatar service is ever contacted, so the content security policy (`img-src 'self' data:`) needs no relaxation.

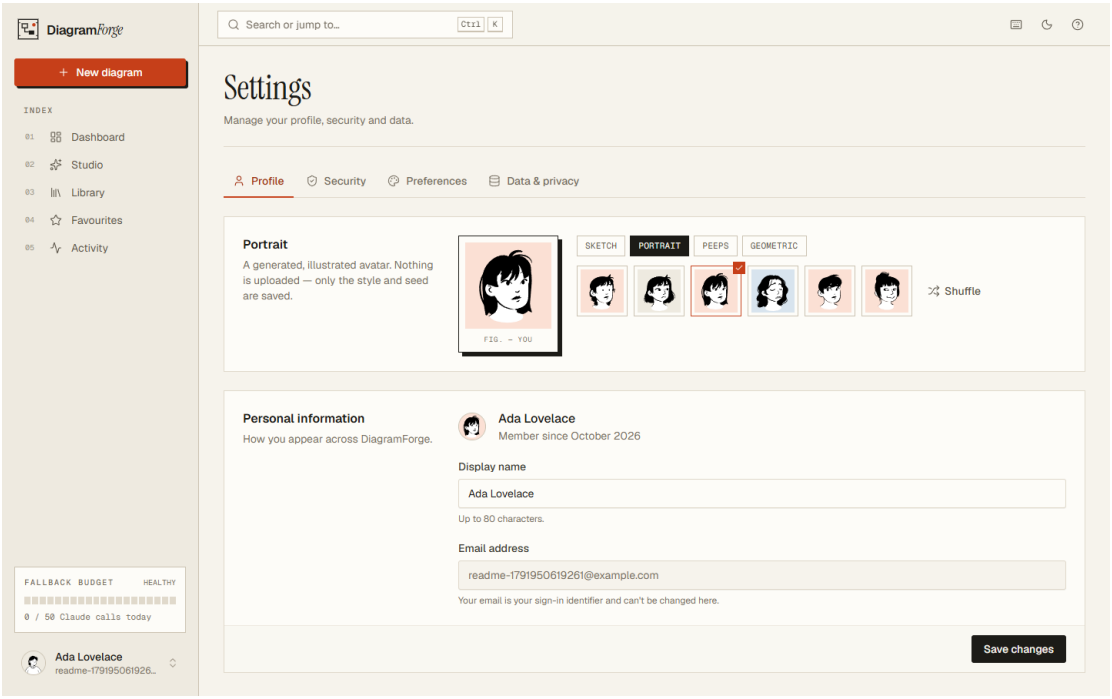


Figure 9: Avatar studio on the profile page, captured from the live cluster

5 DevSecOps Pipeline and Infrastructure

This section is the operational centre of the project: every piece of infrastructure is provisioned as code, and every code change passes an automated, security-gated pipeline before it can reach a running environment.

5.1 Infrastructure as Code

All AWS infrastructure — VPC, security groups, IAM roles, the Jenkins EC2 instance, the EKS cluster and its node groups, and the ECR repositories — is defined in Terraform modules under `infrastructure/`. State is stored remotely in S3 and locked through DynamoDB, so it is safe to share and cannot be corrupted by two concurrent applies. Two deliberate cost/security trade-offs are recorded here rather than hidden:

- **Worker nodes sit in public subnets**, avoiding a NAT Gateway (about $32/month plus data charges). This is a real security trade-off, made for cost and called out explicitly.
- **MongoDB runs as a single-replica StatefulSet**, not a replica set — an honest limitation, recorded for the production-readiness review in Phase 16.

5.2 CI/CD Pipeline Architecture

The pipeline runs on Jenkins as a Multibranch Pipeline job per service. Each DevSecOps gate is implemented once, as a reusable step in a **Jenkins Shared Library** — `sonarQubeScan()`, `owaspDependencyCheck()`, `trivyScan()`, and `pushToECR()` — kept in its own Git repository, because Jenkins can only load a shared library from the root of a repository. Figure 10 shows the backend pipeline; the frontend pipeline is identical apart from the unit-test stage. Any

gate can fail the build independently, a failure skips every later stage, and workspace cleanup (`cleanWs()`) always runs.

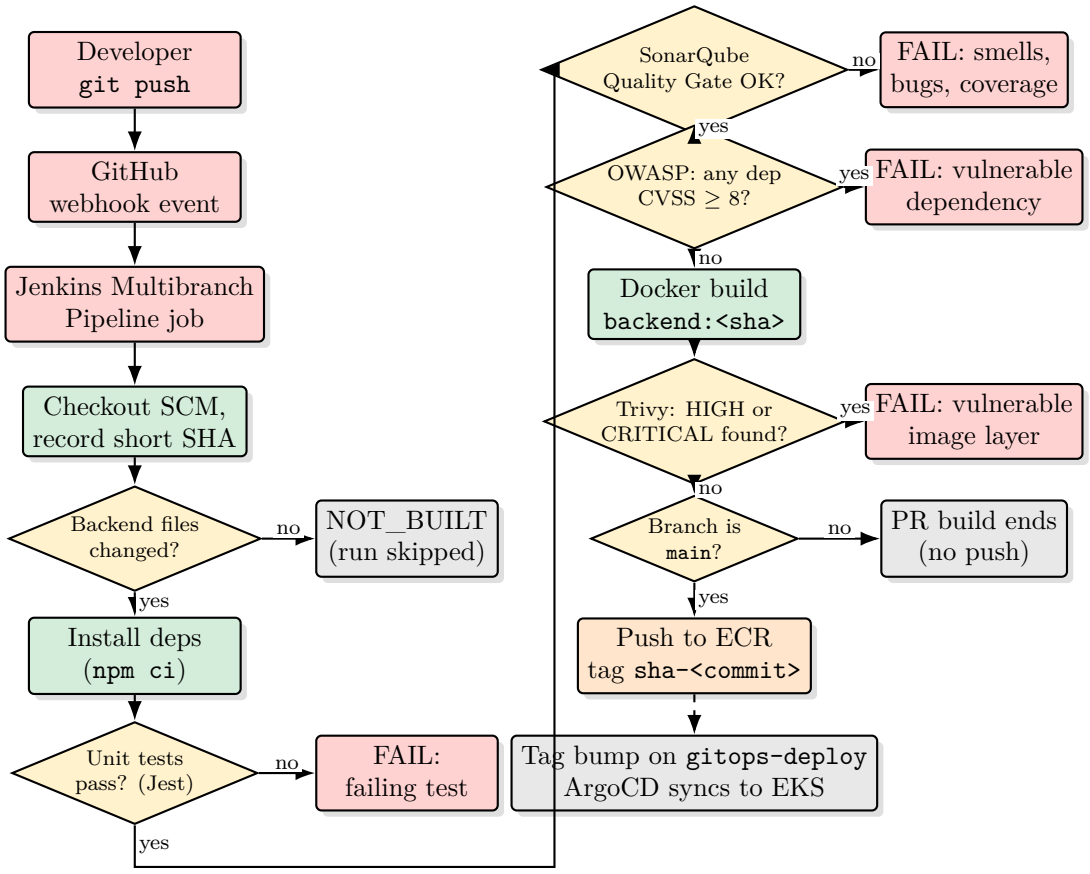


Figure 10: CI/CD pipeline with DevSecOps gates (backend service)

5.3 Security Gates in Detail

Table 4 summarises each gate, the tool behind it, and the requirement it enforces.

Gate	Tool and configuration	What it enforces
Static analysis	SonarQube, via the `withSonarQubeEnv` and `waitForQualityGate` pipeline steps	Bugs, code smells, security hotspots, and coverage thresholds on *new* code. The quality-gate result is delivered back to Jenkins by a SonarQube webhook.
Software composition analysis	OWASP Dependency-Check, `--failOnCVSS 8`, with a persistent local NVD cache	`NFR-S5`: the build fails if any dependency carries a known CVE with CVSS ≥ 8.
Container image scanning	Trivy, `--severity HIGH,CRITICAL --exit-code 1`	`NFR-S5` (image half): the build fails if the built image contains a HIGH or CRITICAL vulnerability, at either the OS-package layer or the language-ecosystem layer.
Registry push	Amazon ECR, immutable tags, tag `sha-<commit>`	Runs only on `main`, and only after every earlier gate has passed. Tagging by Git SHA keeps every running image traceable to the exact source that produced it.

Table 4: DevSecOps gates and what each one enforces

5.3.1 Case Study: A False-Positive CVE from a CPE Vendor Collision

While hardening the OWASP gate, a finding — `CVE-2026-9753`, CVSS v3 base score 8.1 — was reported against the `mongodb` npm driver (version 6.20.0, pulled in transitively by Mongoose). Reading the vulnerability's own affected-software CPE ranges in the raw JSON report, rather than the one-line summary, showed that the CVE actually describes a *MongoDB Server* aggregation-pipeline crash (`$_internalApplyOplogUpdate`), fixed in `mongod` versions 7.0.35, 8.0.24, 8.2.10, and 8.3.3 — different software with an unrelated versioning scheme. NVD's CPE dictionary uses the same vendor/product identifier (`cpe:2.3:a:mongodb:mongodb`) for both the server and the driver, so CPE-based matching flags any project that merely *talks to* a MongoDB server. Upgrading the driver could never clear the finding; this was confirmed directly, since version 6.21.0 was still flagged. The remediation applied is a suppression rule scoped narrowly to `pkg:npm/mongodb@*` and this single CVE ID, with its justification recorded in the suppression file itself — not a blanket suppression, and not a disabled gate.

5.3.2 Case Study: A Real Dependency Vulnerability

The same investigation surfaced a genuine finding: `GHSA-2883-xcg3-v3hh` in `js-yaml` 4.3.1 (pulled in by ESLint), a CPU-exhaustion denial of service triggered by YAML merge keys, fixed in 4.3.2. This one was fixed properly, with an npm `overrides` entry scoped to the ESLint dependency chain only, leaving an unrelated `js-yaml` 3.x used by Jest untouched. The two cases illustrate the distinction the gate is designed to force: a finding is either *fixed* or *proven not to apply*, and the proof is written down.

5.3.3 Case Study: An Unused `npm` CLI as Image Attack Surface

Trivy separately reported four HIGH-severity vulnerabilities in the built backend image, all under `usr/local/lib/node_modules/npm/` — that is, inside the `npm` CLI bundled with the `node:24-alpine` base image, not in the application's own dependencies, which showed zero findings. The container's entrypoint invokes `node` directly and never calls `npm` or `npx` at runtime, so npm was only ever needed for the image's `npm ci` build step. The fix removes npm from the

final image layer once installation completes, which eliminates the unused attack surface rather than merely suppressing the scanner — a stronger remediation, available precisely because the tool was genuinely unnecessary at runtime. It was verified by rebuilding the image, confirming the application still starts and serves `/healthz` with HTTP 200, and re-running Trivy to confirm zero findings.

5.4 Deployment Architecture

Figure 11 shows where each component physically runs once deployment is complete. The Jenkins server, EKS cluster, and ECR repositories, together with the in-cluster workloads and the ALB routing (Phase 7), and GitOps-driven delivery with monitoring (Phases 8–9), were built and verified live (Sections 5.5 and 5.6). A deliberate operational habit accompanies this architecture: the EKS cluster and its load balancer are torn down with `terraform destroy` at the end of each working session and rebuilt in roughly fifteen minutes at the start of the next, because the EKS control plane bills $0.10/hour whether or not any workload is running. Pods pull their images from ECR; the frontend's pull edge is omitted from the figure for readability.

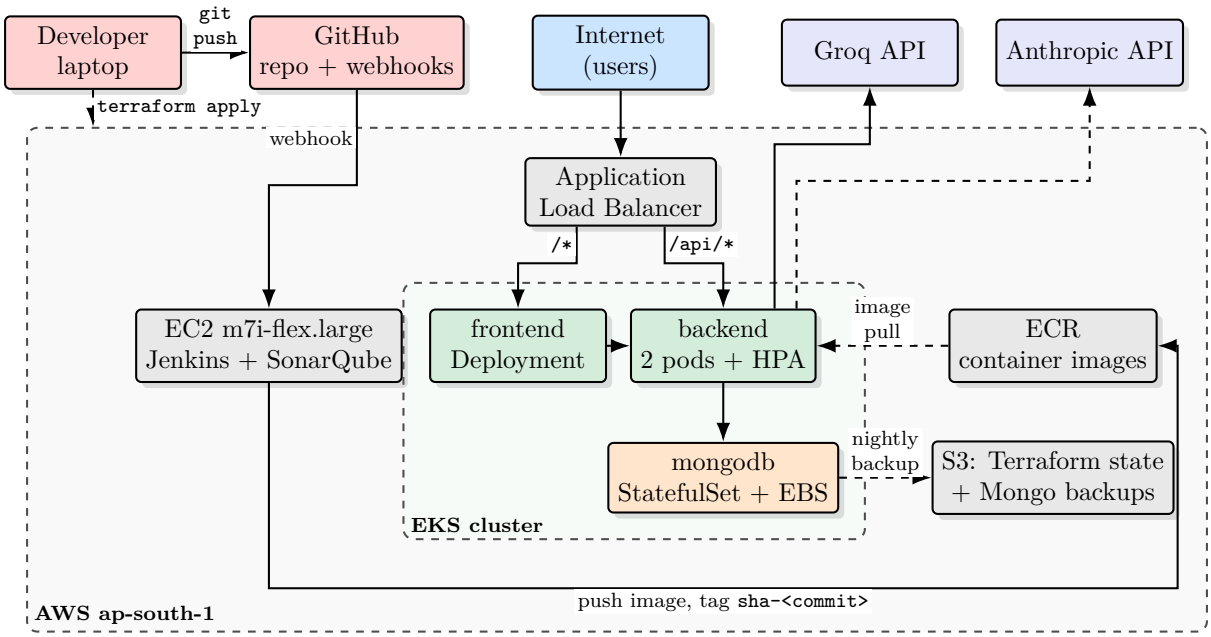


Figure 11: Deployment architecture on AWS

Aspect	Choice	Reasoning
Region	`ap-south-1` (Mumbai)	Lowest latency from India; one region everywhere for consistency.
Worker nodes	Public subnets	Avoids a NAT Gateway (about $32/month); a real, documented security trade-off.
MongoDB	Single-replica StatefulSet	Honest limitation; production would use a replica set or a managed document store.
Jenkins	Single EC2 instance, no HA	It is CI infrastructure, not the product: its downtime blocks deploys, not users.
Backend replicas	Dev 1; prod 2 with an HPA to 6	The production values survive losing one pod with headroom to spare (`NFR-A2`); the dev values keep the verification session cheap.
Secrets	Never in Git or images	Today: supplied at install time from a git-ignored file by the deploy script. Planned (Phase 12): fetched at runtime from AWS Secrets Manager via the External Secrets Operator.

Table 5: Deployment design decisions

5.5 Live Deployment Verification and Lessons Learned

Once the pipeline had pushed both images to ECR, the cluster was created with Terraform and the Helm chart was installed with a single script that reads secrets from a git-ignored file. The deployment was then verified end to end, rather than assumed from a successful `helm install`:

- all pods reached `Ready`; MongoDB's 2 Gi EBS volume was bound through the EBS CSI driver; the Application Load Balancer became `active` and its target `healthy`;
- through the cluster's own reverse proxy, a user was registered and signed in, a diagram was generated by the live primary provider in about 0.6 s (proving the secrets reached the backend), and saved;
- the MongoDB pod was deleted; the StatefulSet recreated it and reattached the *same* persistent volume, and the saved diagram was intact — the concrete justification for a StatefulSet over a Deployment;
- after the session the cluster, nodes, load balancer and volume were destroyed and their absence confirmed in AWS.

The first install did not work, and the reasons are instructive. The chart rendered and installed without error; every failure was a property of the *cluster*, discovered only by running it (Table 6).

Gap	Symptom	Root cause and fix
No default StorageClass	MongoDB pod `Pending`; claim event "no storage class is set"	EKS ships only the legacy in-tree `gp2` class, which is not the default. Added a default `gp3` class on the EBS CSI driver, applied by the deploy script.
Load balancer subnet discovery	Ingress never received an address; controller log: 403 on `DescribeRouteTables`	The hand-copied IAM policy predated the installed controller (v3.6.0). Permission added.
Load balancer creation	Target group created, then 403 on `GetSecurity GroupsForVpc`	Same cause. Rather than patching one action at a time, the file was replaced by upstream's official policy for the installed version: a strict superset, eight actions added.
Backend crash loop	`getaddrinfo ENOTFOUND mongodb`	A symptom, not a defect: a headless Service publishes no DNS records until a pod is ready, so the backend could not resolve a database that never started. Resolved by fixing the storage class.

Table 6: Cluster-level gaps found by the first live deployment

Running the pipeline against real code also surfaced issues that no amount of reading the Jenkinsfile would have revealed:

- **Integration tests need a database.** GitHub Actions supplies MongoDB as a service container; the Jenkins agent had none, so the integration suite timed out. The test stage now starts a throwaway `mongo:7` container on a random localhost port and always removes it.
- **A newly published critical advisory.** OWASP Dependency-Check failed the backend on `proxy-addr` 2.0.7 (CVSS 9.1, IP spoofing via IPv4-mapped IPv6 addresses), published after the previous audit. `npm audit fix` resolved it inside the existing version ranges, with all 77 tests still passing.
- **Operating-system CVEs in the base image.** Trivy flagged HIGH and CRITICAL findings in the frontend image's `musl`, `nghttp2`, `libtiff` and `zlib`, all with fixes available. Both runtime images now run `apk upgrade`; the frontend does so as root and returns to its unprivileged user.
- **An irrelevant analyzer.** Dependency-Check's `pnpm` analyzer failed the stage on an npm-only project; it is disabled in the shared library alongside the already-disabled Yarn analyzer.
- **SonarQube Community has one analysis stream.** Overlapping builds of different branches submitted analyses out of order and the server rejected the later report. Running one analysis at a time, or restricting analysis to `main`, avoids it.
- **No way to rebuild without a code change.** The relevant-changes check inspects only the latest commit, so `main` could not be rebuilt after an infrastructure-only merge. A `FORCE_BUILD` parameter now bypasses the check.

Each of these was diagnosed by reading the actual error rather than the summary line, fixed in code, and merged through a reviewed pull request; none required a manual change to a running system that was not then captured in the repository.

5.6 GitOps and Observability on the Live Cluster

Phases 8 and 9 replaced the hand-run `helm install` with a pull-based model. A single object, the root Application, is applied once; it watches a folder of child Applications in Git, so the application, the monitoring stack and the metrics server all arrive through commits. Jenkins no longer deploys: after a green `main` build its last stage writes the new image tag into the production values on a dedicated `gitops-deploy` branch, which ArgoCD watches, so `main`'s

review rules are untouched. The two pipelines may finish together, so the stage rebases and retries its push.

The stack was brought up on a fresh cluster and checked:

- four ArgoCD applications (root, application, monitoring, metrics server) reached `Synced` and `Healthy`; the application deployed from `gitops-deploy` with the tags Jenkins had recorded and no manual Helm command;
- the backend exposes Prometheus metrics at `/metrics`, a request-latency histogram labelled by route pattern so cardinality stays bounded; the ingress forwards only to the frontend, so the endpoint is not reachable from the internet. Prometheus scraped both pods;
- the four alert rules (5xx rate, p95 latency, no backend available, MongoDB volume nearly full) loaded without errors, and the Grafana dashboard defined in the chart was picked up automatically by the Grafana sidecar (Figure 12);
- secrets stay out of Git: the application Secret is created out of band by the bootstrap script from a git-ignored file, and the chart's `secrets.create` switch tells ArgoCD not to own it.



Figure 12: Grafana dashboard provisioned by the Helm chart, captured live: request rate by status, latency percentiles, and CPU and memory per backend pod

Again, the chart and manifests were correct on paper and two properties of the cluster were found only by running it, plus a set of pipeline problems found only by running the new stage for real (Table 7):

Gap	Symptom	Root cause and fix
Subnets not labelled for load balancers	Ingress without an address; controller: "subnets count less than minimal required"	The public subnets lacked the `kubernetes.io/role/elb` tag, a step done by hand in the previous session and never recorded. Now tagged in Terraform; the plan is a no-op on the live cluster.
No metrics server	HPA showed `cpu: <unknown>`; ArgoCD reported the application `Degraded`	EKS does not ship one. Added as an ArgoCD application; the HPA then read the real CPU figure and the application turned `Healthy`.
SonarQube coverage gate	Gate failed on new-code coverage at 0%	The new `/metrics` code was the first genuinely new code the gate saw, and no coverage report was imported. Jest now writes lcov and `sonar-project.properties` points the scanner at it.
Immutable ECR tags	Rebuilding a commit failed with "tag is immutable" and skipped later stages	The push step now treats only that refusal as success; the Jenkins role cannot describe images, so an existence check was not an option.
Token scope	403 pushing the `gitops-deploy` branch	The Jenkins GitHub token had read-only contents access; permission raised without changing the token.
Narrow clone refspec	Second pipeline's push rejected as non-fast-forward	Jenkins fetches only the branch under test, so `origin/gitops-deploy` never resolved and each run restarted from `main`. Explicit refspecs fixed the lookup.

Table 7: Problems found by running the GitOps stack for real

Two security improvements came with the same work: the NVD and SonarQube tokens are no longer passed as command-line arguments (where the process list would show them) but through a private temporary properties file and the scanner's environment variable respectively. One property was not demonstrated: that ArgoCD reverts a manual change (self-heal) was configured but the test was not run in this session.

6 Implementation Status

6.1 Phase Roadmap

The project is executed as incremental phases, each with one concrete learning outcome (Table 8). This report reflects the system after Phase 9, with Phases 0–9 complete and verified live.

Phase	Focus	Main learning outcome
0–1	Foundations, application	React, Express, MongoDB, Mermaid, JWT auth, multi-provider LLM with fallback, Docker Compose
2	Jenkins infrastructure	Terraform modules, EC2, VPC, IAM, remote state
3	Jenkins configuration	Configuration as Code, credentials, SonarQube toolchain
4	EKS cluster	Kubernetes on AWS, node groups, IRSA, ALB controller
5	ECR	Private registries, immutable tags, lifecycle policies
6	CI gates	Jenkins pipelines, tests, SonarQube, OWASP Dependency-Check, Trivy (complete; all gates passing)
7	Helm	Templated manifests, probes, HPA, StatefulSet (complete; verified live on EKS)
8	GitOps	ArgoCD App-of-Apps, auto-sync, self-healing, tag bumps by the pipeline (complete; verified live)
9	**Observability**	**Prometheus, Grafana, application metrics, alerts (complete; verified live)**
10	DNS and TLS	Route53, ACM certificates, HTTPS through the ALB
11	Persistence and backup	EBS CSI, S3 backups, restore testing
12	Security hardening	External Secrets, NetworkPolicy, RBAC, Pod Security Standards
13	Supply chain and policy	SBOM, image signing, admission policies, IaC scanning
14	Progressive delivery	Argo Rollouts, canary releases, metric-based rollback
15	Logs, traces, incidents	Loki, OpenTelemetry, runbooks, SLOs
16	FinOps and resilience	Budgets, Karpenter, chaos tests, disaster recovery

Table 8: Build roadmap

6.2 Verified Outcomes

The following have been directly verified, not merely designed:

- 77 automated Jest tests across 9 suites (eight unit suites and one integration suite that runs against a real MongoDB instance), passing both in GitHub Actions and in the Jenkins pipeline, plus real-browser end-to-end runs of the frontend in light, dark and mobile layouts.
- A passing SonarQube Quality Gate on the backend project, with the result delivered back to Jenkins by webhook.
- A clean OWASP Dependency-Check run (exit code 0) after the remediations described above, completing in under two minutes once analyzers irrelevant to an npm-only project (OSS Index, Maven Central, Yarn audit) were disabled — previously over thirty minutes.
- A clean Trivy scan of the backend image (zero findings) after removing the unused npm CLI.
- Terraform-managed infrastructure with remote state (S3 plus DynamoDB locking), applied and destroyed repeatedly as part of the project's cost discipline.
- Both container images built, scanned and pushed to ECR by the Jenkins pipeline on `main`, each tagged with its commit SHA.
- A live deployment to Amazon EKS through the Helm chart, verified end to end including persistence across deletion of the database pod (Section 5.5), then destroyed.
- GitOps delivery through ArgoCD and live monitoring with Prometheus and Grafana on the same cluster, verified and then destroyed (Section 5.6).

7 Security Considerations

- **Secrets never enter version control.** CI secrets (GitHub token, SonarQube token, Anthropic API key, NVD API key) are stored as Jenkins credentials, injected with `withCredentials`, and masked in build logs. A committed `.env.example` documents variable *names* and placeholders only.
- **Least-privilege file permissions.** The server-side secrets file is locked to `600 root:root`; a world-readable copy found during a routine audit was corrected as part of this work.
- **A known residual risk.** Jenkins masks credentials in build *logs*, but a secret passed as a command-line argument remains visible in the OS process list to anyone with shell access on the build host. The NVD key is currently passed this way; moving it to an environment variable read by the tool is a recorded follow-up.
- **IAM permission additivity.** A resource-based policy, such as an ECR repository policy, can only *add* permission; it cannot subtract what a principal's identity policy already grants. True exclusivity requires an explicit `Deny`.
- **Ownership checks return `404`, not `403`.** Confirming that a resource exists but is forbidden would let an attacker enumerate valid IDs.
- **Password handling.** bcrypt with a cost factor of 12; `User.toJSON()` strips `passwordHash` during serialisation, so the hash cannot leak even through a careless `res.json(user)`.
- **No static cloud keys in the cluster.** The load balancer controller and the EBS driver obtain temporary AWS credentials through IAM Roles for Service Accounts, each role trusted by exactly one Kubernetes service account. The controller's policy is the upstream policy for its installed version, not a hand-maintained copy.
- **Known gaps, scheduled rather than hidden.** Kubernetes Secrets are base64, not encryption; there is no NetworkPolicy or Pod Security enforcement yet; the in-cluster MongoDB is a single member without authentication; and the load balancer serves plain HTTP with no custom domain. These are addressed in Phases 10–12.

8 Cost Model

Table 9 shows monthly AWS cost in `ap-south-1`, running continuously versus only during active working sessions.

Resource	Always on	About 15 h/week
EKS control plane ($0.10/h, billed even when idle)	$73	pro-rated
2 × m7i-flex.large worker nodes (spot)	$60 on demand, less on spot	pro-rated
Jenkins EC2 (m7i-flex.large)	$70	stopped when idle
Application Load Balancer	$18 + data	pro-rated
EBS volumes	$5	$5 (persists)
Route53 hosted zone	$0.50	$0.50 (persists)
ECR storage	negligible	negligible
Total	**about $215/month**	**about $25–35/month**

Table 9: AWS cost, always on versus disciplined teardown

The affordability strategy is enforced by habit rather than tooling: destroy the EKS cluster

and stop the Jenkins instance at the end of every session. That is only survivable because the whole cluster is code — rebuilding it by hand through the AWS console, repeatedly, would be impractical. This is automated by two scripts: one installs the application onto a fresh cluster and one removes the application, its load balancer and volume, and then the cluster itself, in the order that avoids orphaned resources. Worker nodes run on spot capacity in public subnets, so no NAT gateway (about $32/month) is ever created. AWS Budgets alerts are configured at $25, $50 and $75 of monthly spend. While idle between sessions the account costs roughly $7 per month (two stopped instances' disks and one public address); everything else is destroyed.

9 Related Work

This project sits at the intersection of two established practices rather than proposing a new technique in either. **GitOps** — declarative desired state in Git, continuously reconciled by an in-cluster controller — is the operating model popularised by tools such as ArgoCD and Flux, and is used here to keep human operators out of the direct `kubectl` path. **DevSecOps** — shifting security scanning left into the CI pipeline instead of treating it as a post-deployment audit — is realised through three independent, build-failing gates rather than one monolithic check, so that every failure is attributable to a specific class of defect. The case studies in Section 5 are offered as concrete, reproducible illustrations of a failure mode that is common but rarely documented in detail: CPE-based vulnerability matching that conflates unrelated products sharing a vendor/product name in the NVD dictionary.

10 Future Work

The remaining phases (Table 8) extend the platform toward a production-representative posture:

- **Alert routing** (extends Phase 9): the four alert rules evaluate correctly, but Alertmanager has no receiver yet; routing to a channel and tying thresholds to the SLOs in Table 3 follows with Phase 15.
- **TLS, backup, and hardening** (Phases 10–12): end-to-end HTTPS, restore-tested backups, Pod Security Standards, and NetworkPolicy enforcement.
- **Supply-chain security** (Phase 13): image signing with Cosign and admission-time policy with Kyverno, so the cluster refuses any image the pipeline did not sign — closing the loop that the current gates open.
- **Progressive delivery and resilience** (Phases 14–16): canary releases with metric-based rollback, chaos testing, and FinOps controls.

11 Conclusion

DiagramForge shows that an instructive DevOps project does not need a complex application — it needs a complete one, wrapped in a platform that is not simplified for the sake of the exercise. The application layer is intentionally small; the pipeline, infrastructure, and security posture around it are not. The case studies in Section 5 were all found by running the gates against real code and reading the underlying report data rather than trusting a summary line, and each was resolved with a documented, individually justified fix. The first live deployment repeated the lesson at cluster scale: a chart that linted and installed cleanly still failed on three properties of the cluster, and each was found by reading the controller's own error and fixed in code. Bringing up the GitOps stack repeated it a third time (unlabelled subnets, an absent metrics server, a read-only token). That discipline, more than any single tool choice, is the real subject of this report.

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